Group 19 – Project Proposal for Predicting a Patient's Length of Stay

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A comparative study of different Machine Learning models to predict Patient's Length of stay. In this research, we aim to build and propose a machine learning framework for predicting a patient's length of stay based on different data factors. We will be evaluating different frameworks against each other and proposing the best one.

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With varying demand but a fixed capacity of beds, bed managers at NHS Wrightington, Wigan and Leigh (WWL) have the difficult job of making critical decisions in facilitating the effective use of acute hospital beds, often relying on their own experience, initiative, supplementary data, and forecasts during this process. In recent times, tools taking advantage of artificial intelligence and other statistical methods have been developed with the objective of assisting bed managers in making these decisions. One such method is predicting an individual patient's length of stay – the period from admittance to a ward to discharge from the hospital. Predicting a patient's length of stay can conceivably be completed by considering a range of patient information, including their demographics, diagnoses, and comorbidities.

CCS CONCEPTS • Machine Learning • Artificial Intelligence • Health Informatics

1 INTRODUCTION

A patient's Hospital length of stay (LOS) is defined as the time and number of days an inpatient is admitted to the hospital and is using the bed. The LOS is defined for a single admission event; that is, how many consecutive days has the patient been in the hospital? This can vary from anytime between 0 to 50+ days and depends on a number of factors that will be discussed further.

Determining a patient's LOS will help the hospitals assign better resources and establish appropriate healthcare planning by providing medical facilities and personnel accordingly. This not only helps manage the resources effectively but also takes the strain off healthcare workers without overworking them. Healthcare systems are becoming more cost-conscious, and having concrete data along with forecasting systems in place can help reduce the associated cost and improve overall patient care. Similarly, it may also help understand the underlying patterns in the data, especially which wards and hospital care units experience the most requirement for beds, which can significantly boost the evaluation of operational functions of a hospital. [1]

Alternatively, from a patient's perspective, shortening their LOS is an ideal scenario since it mitigates the dangers of them remaining in care for longer than required. These include falls, hospital-acquired infections and medication errors, all of which could in turn threaten a patient's life and lead to a prolonged LOS, straining the hospital's resources even more. [1]

Many methods have been developed for determining the LOS, and while some literature focuses on determining LOS through a pre-defined disease group, treatment, or medication [2], it involves many complicated factors, including but not limited to their characteristics, i.e., their ethnicity, pre-existing complications, or general discharge planning.

2 RESEARCH QUESTION AND OBJECTIVES

Goal	Research Question / Objective	Measurable Output
Primary (Operational)	How effectively can a predictive model, utilising pre-admission and early-spell clinical and operational data, accurately forecast the patient's LOS spell_episode_los to support proactive bed management and resource optimisation?	Model performance metrics (e.g., RMSE, MAE, ROC-AUC) on a held-out test set.
Secondary (Interpretability)	To identify which clinical and operational features are the most important predictors of a patient's prolonged or short Length of Stay.	Feature importance analysis and ranking of variables.

3 DATA UNDERSTANDING

Our project was conducted in collaboration with the NHS at Wigan, Wrightington and Leigh (WWL). The dataset provided contains 41,846 rows and 101 columns, comprising 56 numerical and 45 categorical variables. These include patient demographics, admission and discharge timestamps, ward codes, high-level diagnosis and procedure codes, comorbidity indicators, COVID-19 flags, specialty information, and measures of deprivation.

The target variable, spell_episode_los, represents the patient's length of stay in days and is computed as the difference between discharge and admission dates. Exploratory analysis shows a mean LOS of 1.76 days, a median of 0 days, a standard deviation of 5.28, and a maximum LOS of 123 days, indicating that most patient spells are very short with a long right-skewed tail of extended admissions.

The dataset also contains a high proportion of missing values, particularly in variables where the probability of a recorded event is naturally low (e.g., covid_19_diagnosis_flag). This required careful handling and informed the preprocessing strategies used in the project.

Correlation analysis highlighted substantial distribution issues in some key predictors. For example, **duration_elective_wait** was extremely right-skewed (figure 1), with wait times ranging up to 3,000 days, yet its median was –1 (representing emergency cases). Despite this wide spread, its correlation with **spell_episode_los** was surprisingly weak and slightly negative ($r \approx -0.07$). This suggests that patients with long planned waits typically undergo routine, well-managed procedures with short, predictable stays, while emergency admissions—those with zero wait time—drive most of the variability in LoS. These differences further justify using non-linear, tree-based models such as XGBoost, which can better separate these patient groups than linear methods.

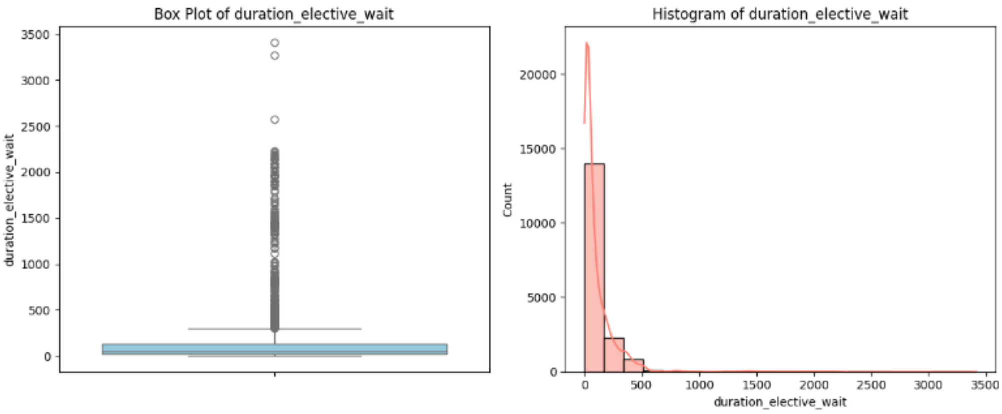

Figure 1 (Histogram)

4 DATA PRE-PROCESSING

4.1 Data Leakage Prevention and Feature Removal

To ensure that models only use information realistically available at or near the time of admission, a substantial list of columns was removed to prevent data leakage.
These included:

- Post-admission date fields such as discharge_date_dt and discharge_created_datetime_dt.
- Outcome columns such as delayed_discharges_no_of_days, delayed_discharges_flag, discharge_delay_reason_description, medically_optimised, and patient_age_on_discharge.
- Redundant or high-detail text fields, including spell_primary_diagnosis_description, ward_name_discharge, and similar descriptive columns.

4.2. Data Cleaning and Imputation Strategy

We apply a domain-aware strategy for missing data Key elements include:

1. **Numerical Imputation:**
   Several numerical fields are imputed with a placeholder value of -1. In this dataset, missing values often indicate "not recorded" or "not applicable", instead of missingness at random, so preserving this signal allows the model to learn from the absence of information. This applies to variables such as NEWS2, duration_elective_wait, and acuity_code.

2. **Categorical Imputation:**
   Missing categorical features are mapped to explicit placeholder categories such as "Not Specified", "Unknown Location", or "Not Applicable", ensuring the model retains this information rather than treating missingness as noise.

3. **Frailty Score Cleaning:**
   The frailty_score column contained a mix of numerical and text values (e.g., 2 and 2 – healthy), so we simply extracted the numerical values from the text to keep the values consistent throughout.

4.3. Encoding and Scaling

Because the dataset contains highly heterogeneous feature types, different preprocessing techniques were applied:

1. **Target Encoding:**
   High-cardinality categorical features (almost 21 columns) — including specialty_spec_code, hrg_group, and various clinical and geographical codes — were encoded using a Target Encoder. This converts each category into a single numerical value representing the mean length of stay (or class

probability for the classifier). This approach avoids creating multiple dimensions with one-hot encoding or accidental ordinality that comes with label encoding and captures meaningful category–target relationships.

2. **Standard Scaling:**
   Variables that show approximately normal distributions, such as patient_age_on_admission and deprivation_decile, were standardized using Z-score scaling.

3. **Robust Scaling:**
   Features with heavy tails or outliers, including duration_elective_wait and comorbidity_score, were scaled using a Robust Scaler, which uses the median and interquartile range and is therefore less sensitive to extreme values.

5 MODEL STRATEGY

Exploratory analysis confirmed that the target variable, **spell_episode_los,** is extremely right-skewed. Key statistics:

- Mean LOS: 1.76 days
- Median length of stay: 0 days
- Maximum length of stay: 123 days

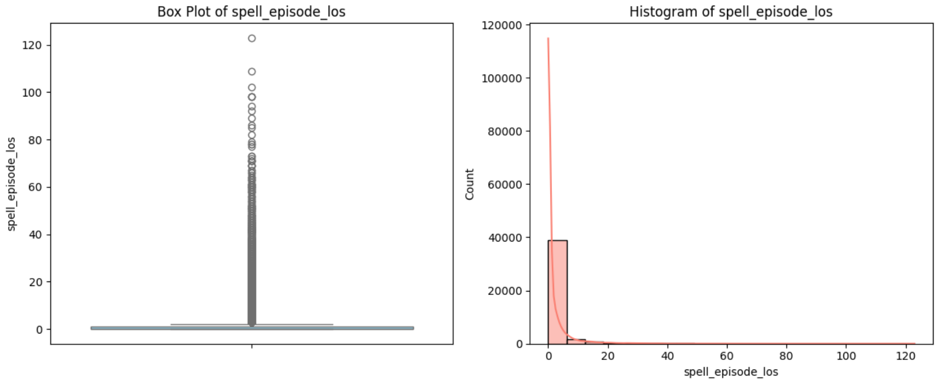

Figure 2 Target Variable Histogram

This supported two decisions:

1. **Log Transformation:** We use log(1 + LOS) to reduce skew and stabilise variance. When calculating the final LOS, we can reverse it back to the original values, since the transformation is reversible.
2. **Specialized Model Strategy:**
   A dedicated outlier classifier and tailored preprocessing were used to better model long-stay cases.

5.1 Modelling Methodology and Results

Two modelling flows were evaluated to address the extreme right skew in *spell_episode_los*. Flow 1 applies a single regression model across the full dataset, while Flow 2 uses a two-stage hybrid approach that explicitly isolates long-stay outliers before modelling LOS.

6 Model Flow Evaluation

6.1 Flow 1: Single Regression on Full Dataset

In this approach, LOS prediction is treated as a regression problem using the log-transformed target log (1 + LOS)

Four models were trained with hyperparameters selected via cross-validation:

- XGBoost Regressor
- Random Forest Regressor
- Ridge Regression
- LASSO Regression
- XGBoost-Poisson

Best Model: **XGBoost Regressor**

Metric	Value
RMSE	0.4816
MAE	0.2443
R^2	0.6667

Although performance is reasonable, residual analysis showed that a single model struggles to simultaneously capture both the dense cluster of short stays and the small group of long-stay patients. This motivated the development of Flow 2. The compared scores of all models:

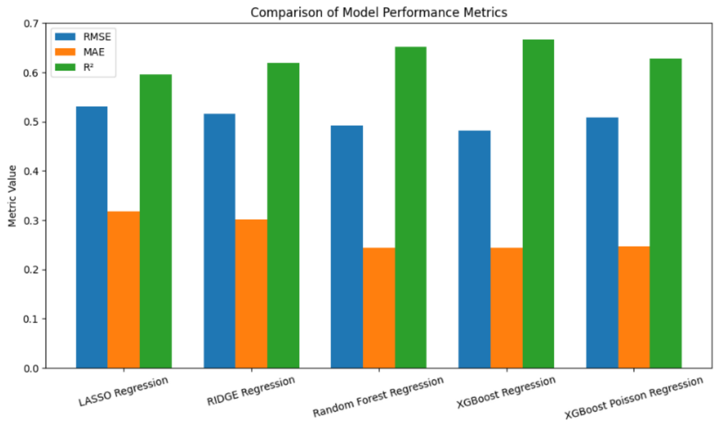

Figure 3 Flow 1 Regression Metrics

6.2 Flow 2: Two-Stage Hybrid Approach

Flow 2 separates common inlier cases from long-stay outliers to improve predictive accuracy for the majority of patients.

6.2.1 Stage 1: Outlier Classification

Outliers were defined using the 1.5 × IQR rule applied to raw LOS. A Random Forest and a Logistic Regression classifier were trained using only features available at admission.

Model	**ROC–AUC Score**
Random Forest Classifier	0.9361
Logistic Regression Classifier	0.9372

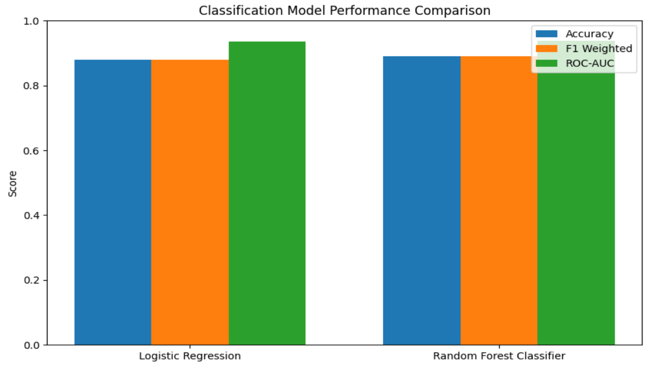

Figure 4 Classification Metrics

Both models demonstrate a strong ability to identify long-stay risk early in a patient's spell.

6.2.2 **Stage 2: Inlier and Outlier Regression**

Separate regression models were trained on the log-transformed LOS for inliers and outliers. The average errors found across all models can be found in the table below:

Model	**Data Type**	**RMSE**	**MAE**	**R^2**
Inlier Regression	Normal (non-outlier) stays	≈ 0.22	≈ 0.10	≈ 0.58
Outlier Regression	Long-stay outliers	≈ 0.58	≈ 0.44	≈ 0.19

The inlier model substantially improves predictive accuracy compared with Flow 1. Performance on extreme outliers remains limited due to the unpredictable nature of complex social and clinical delays. The performance for the Inlier Models vs Outlier Models can be found here:

Figure 5 Inlier Metrics

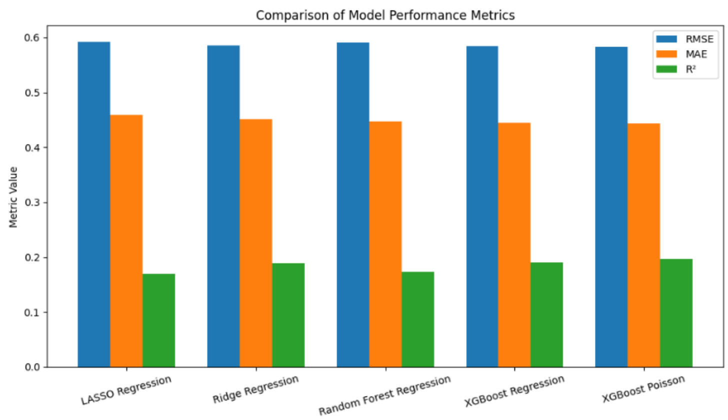

Figure 6 Outlier Metrics

7 FEATURE IMPORTANCE

Random Forest feature importance identified clinically consistent predictors of LOS.

Top Predictors	Reason for Importance
Age on admission	Older patients typically require longer recovery.
Diagnosis groups	Strong determinant of treatment needs and resource intensity.
Comorbidity score	Higher clinical burden increases LOS.
Elective waiting duration	Reflects illness severity and pre-admission delays.
Deprivation indicators	Capture socioeconomic complexity affecting discharge readiness.

These findings confirm that LOS is shaped by clinical complexity, demographic factors, and social/operational conditions.

8 EVALUATION

For the regression models the following metrics were chosen to evaluate our models

- **Root Mean Square Error (RMSE):** This was selected as the primary evaluation metric because it heavily penalizes larger errors (due to squaring the difference). In a medical setting where precision in predicting short, high-turnover stays and minimizing errors on crucial long stays are vital, the RMSE forces the model to maintain high precision.
- **Mean Absolute Error (MAE):** This metric provides the average magnitude of the errors, giving an easily interpretable measure of the typical prediction error (e.g., "the model is off by X days on average").
- **R^2 (Coefficient of Determination):** Used to determine the proportion of the variance in the target variable (LoS) that is predictable from the features. This metric is used to compare the overall explanatory power across different models.

For the binary classification model (used in Flow 2 to distinguish between **Normal** and **Outlier** patients), we focused on the **ROC-AUC (Receiver Operating Characteristic - Area Under the Curve)**. ROC-AUC measures the model's ability to distinguish between the two classes across all possible classification thresholds.

9 RESULTS INTERPRETATION

Flow 2 is clearly the stronger operational strategy for predicting Length of Stay. It cuts the error for most routine admissions by about half, leading to noticeably lower MAE and RMSE for the patients who make up the bulk of hospital activity. While both flows show similar explanatory power (e.g., comparable R^2), Flow 2 is far better calibrated for short stays—crucial for day-to-day bed management. By explicitly modelling long-stay outliers, it can identify potential long-stay patients much earlier, enabling proactive planning for scarce resources. This combination of better accuracy for routine cases and targeted handling of outliers makes Flow 2 a far more robust and operationally useful approach.

9.2 Addressing the Research Questions

Primary Question: Resource Optimisation

The modelling framework strongly supports resource optimisation.

- Risk Flagging: The classifier in flow 2 achieves ROC–AUC ≈ 0.94, enabling early identification of long-stay patients. This can help in early intervention of patients who are at a high risk and may have longer hospital stays.
- Operational Confidence: The inlier model reliably predicts LOS for ~89% of cases, supporting bed planning, staffing, and discharge workflows. For patients, who are not high risk, this can be used to predict their length of stay effectively allowing hospitals to provide resources accordingly.

Secondary Question: Key Influencing Features

Feature importance analysis confirms the dominant influence of age, diagnosis categories, comorbidity, elective waiting duration, and deprivation — all consistent with clinical intuition.

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10 LIMITATIONS AND FUTURE IMPROVEMENTS

The hybrid design was essential for addressing the extreme right skew in LOS. It provides strong accuracy for routine admissions and early warning for outliers — the two most operationally valuable outputs. However, there's still a lot of limitations to the model.

Limitations

1. Prediction Difficulty due to Skew: Due to the severe right skew of the data, predicting the exact duration of high-volume shorter stays is significantly easier than predicting the precise length of rare, complex, long-stay events. The precision gained on the routine cohort is often offset by large errors on these high-leverage outliers, limiting overall model accuracy.
2. Outlier Prediction Noise: Extreme LoS events are often driven by social and clinical complexities, such as care-home placement delays or family factors, which are typically not captured in standard structured electronic health record (EHR) data.
3. Regression-Centric Output: The current design focuses on a point-estimate regression output (an exact number of days). However, operational teams, such as ward managers, may benefit more from predictive LoS categories (e.g., Short, Medium, Long) rather than a precise day count.
4. While train and test split was stratified based on outlier class, a good methodology would be to treat it as time-series data rather than normal one. We could split the data into different quarters and then create observations.

Future Improvements

1. Reframing LoS as Classification: To mitigate the high variance and difficulty in accurately predicting the exact duration of long stays, the problem should be refactored into a classification challenge. This involves predicting which clinically meaningful duration band (e.g., 2 days, 3-7 days, 7 days) a patient will fall into. This shifts the model's objective from maximizing numerical precision to maximizing operational actionability.
2. Incorporating Contextual Data: Future work should explore methodologies to integrate other data sources (e.g., manually logged free-text notes or specific administrative logs for discharge delays) to capture the social and clinical nuance that drives extreme outlier stays.
3. Currently there's not much work that reports how the nurses or other medical staff influence the predictions of a patient's LOS; this could offer a different perspective on assessing LOS as it's been observed that nurses spend more time with patients than physicians [3].

Contributions Sheet

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Project Title : Predicting Patient Length of Stay

Project Host : NHS WWL

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Kumara Guru Kumar	37011327	Data modeling, Predictive analysis	15
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Confirmation of Agreement

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